EXPERIMENT 05

Python Programs — Code and Output Only

1. type() Function

Code

```
a = 25
b = 12.5
c = "Python"
d = True

print("Value:", a, "Type:", type(a))
print("Value:", b, "Type:", type(b))
print("Value:", c, "Type:", type(c))
print("Value:", d, "Type:", type(d))
```

Output (sample)

```
Value: 25 Type: <class 'int'>
Value: 12.5 Type: <class 'float'>
Value: Python Type: <class 'str'>
Value: True Type: <class 'bool'>
```

2. Data Conversion Functions

Code

```
a = "25"
b = "12.5"
c = 100

x = int(a)
y = float(b)
z = str(c)

print("Integer:", x, type(x))
print("Float:", y, type(y))
print("String:", z, type(z))
print("Boolean:", bool(c))
```

Output (sample)

```
Integer: 25 <class 'int'>
Float: 12.5 <class 'float'>
String: 100 <class 'str'>
Boolean: True
```

3. Implicit and Explicit Type Conversion

Code

```
# Implicit type conversion
a = 10
b = 5.5
c = a + b

print("Implicit Conversion:")
print("Result:", c)
print("Type:", type(c))

# Explicit type conversion
x = "20"
y = int(x)

print("\nExplicit Conversion:")
print("Value:", y)
print("Type:", type(y))
```

Output (sample)

```
Implicit Conversion:
Result: 15.5
Type: <class 'float'>

Explicit Conversion:
Value: 20
Type: <class 'int'>
```

4. Conversion of Input Data

Code

```
num1 = int(input("Enter first integer: "))
num2 = float(input("Enter second number: "))

print("First number:", num1)
print("Type:", type(num1))
print("Second number:", num2)
print("Type:", type(num2))
print("Sum:", num1 + num2)
```

Output (sample)

```
Enter first integer: 10
Enter second number: 5.5
First number: 10
Type: <class 'int'>
Second number: 5.5
Type: <class 'float'>
Sum: 15.5
```

5. Mathematical Built-in Functions

Code

```
a = -25
b = 4.5678
numbers = [10, 25, 5, 40, 15]

print("Absolute value:", abs(a))
print("Power:", pow(2, 3))
print("Rounded value:", round(b, 2))
print("Maximum:", max(numbers))
print("Minimum:", min(numbers))
print("Sum:", sum(numbers))
```

Output (sample)

```
Absolute value: 25
Power: 8
Rounded value: 4.57
Maximum: 40
Minimum: 5
Sum: 95
```

6. len(), sum(), max() and min() — Total and Average

Code

```
marks = [75, 82, 68, 91, 85]

total = sum(marks)
count = len(marks)
highest = max(marks)
lowest = min(marks)
average = total / count

print("Marks:", marks)
print("Total:", total)
print("Number of Subjects:", count)
print("Highest Marks:", highest)
print("Lowest Marks:", lowest)
print("Average:", average)
```

Output (sample)

```
Marks: [75, 82, 68, 91, 85]
Total: 401
Number of Subjects: 5
Highest Marks: 91
Lowest Marks: 68
Average: 80.2
```

7. Functions from math Module

Code

```
import math

n = 25

print("Square Root:", math.sqrt(n))
print("Power:", math.pow(2, 3))
print("Ceiling:", math.ceil(4.3))
print("Floor:", math.floor(4.8))
print("Factorial:", math.factorial(5))
print("GCD:", math.gcd(24, 36))
print("Value of Pi:", math.pi)
```

Output (sample)

```
Square Root: 5.0
Power: 8.0
Ceiling: 5
Floor: 4
Factorial: 120
GCD: 12
Value of Pi: 3.141592653589793
```

8. Simple User-Defined Function

Code

```
def greet():
    print("Welcome to Python")

greet()
```

Output (sample)

```
Welcome to Python
```

9. Function Calling

Code

```
def display():
    print("MIT World Peace University")

display()
display()
```

Output (sample)

```
MIT World Peace University
MIT World Peace University
```

10. Function with Minimum Two Arguments

Code

```
def addition(a, b):
    result = a + b
    print("Addition:", result)

addition(10, 20)
addition(25, 15)
```

Output (sample)

```
Addition: 30
Addition: 40
```

11. Positional Arguments

Code

```
def student(name, roll_no):
    print("Name:", name)
    print("Roll No:", roll_no)

student("Rahul", 101)
```

Output (sample)

```
Name: Rahul
Roll No: 101
```

12. Keyword Arguments

Code

```
def student(name, roll_no):
    print("Name:", name)
    print("Roll No:", roll_no)

student(roll_no=102, name="Amit")
```

Output (sample)

```
Name: Amit
Roll No: 102
```

13. Default Arguments

Code

```
def greet(name, message="Welcome to Python"):
    print(name, "-", message)

greet("Rahul")
greet("Amit", "Good Morning")
```

Output (sample)

```
Rahul - Welcome to Python
Amit - Good Morning
```

14. Function with Return Value

Code

```
def addition(a, b):
    result = a + b
    return result

answer = addition(25, 15)
print("Result:", answer)
```

Output (sample)

```
Result: 40
```

15. Returning Multiple Values

Code

```
def calculate(a, b):
    total = a + b
    difference = a - b
    product = a * b
    return total, difference, product

x, y, z = calculate(20, 10)

print("Total:", x)
print("Difference:", y)
print("Product:", z)
```

Output (sample)

```
Total: 30
Difference: 10
Product: 200
```

16. Local Variable

Code

```
def calculate():
    x = 10
    y = 20
    print("Sum:", x + y)

calculate()
```

Output (sample)

```
Sum: 30
```

17. Global Variable

Code

```
x = 100

def display():
    print("Value of x inside function:", x)

display()
print("Value of x outside function:", x)
```

Output (sample)

```
Value of x inside function: 100
Value of x outside function: 100
```

18. Local and Global Variables with Same Name

Code

```
x = 100

def display():
    x = 50
    print("Inside function:", x)

display()
print("Outside function:", x)
```

Output (sample)

```
Inside function: 50
Outside function: 100
```

19. global Keyword

Code

```
x = 10

def update():
    global x
    x = 50

print("Before function call:", x)
update()
print("After function call:", x)
```

Output (sample)

```
Before function call: 10
After function call: 50
```

20. Practical Application — Student Result

Code

```
def calculate_result(m1, m2, m3):
    total = m1 + m2 + m3
    percentage = total / 3
    return total, percentage

m1 = int(input("Enter Python marks: "))
m2 = int(input("Enter Data Structure marks: "))
m3 = int(input("Enter Database marks: "))

total, percentage = calculate_result(m1, m2, m3)

print("Total Marks:", total)
print("Percentage:", percentage)
```

Output (sample)

```
Enter Python marks: 80
Enter Data Structure marks: 75
Enter Database marks: 85
Total Marks: 240
Percentage: 80.0
```

21. Practical Application — Largest of Three Numbers

Code

```
def largest(a, b, c):
    return max(a, b, c)

x = int(input("Enter first number: "))
y = int(input("Enter second number: "))
z = int(input("Enter third number: "))

result = largest(x, y, z)
print("Largest Number:", result)
```

Output (sample)

```
Enter first number: 12
Enter second number: 25
Enter third number: 18
Largest Number: 25
```